Abstract

This project presents the design and simulation of an 8-bit signed binary number representation system utilizing the sign-magnitude method. In digital computing, representing negative numbers is essential for arithmetic operations. The sign-magnitude format allocates the most significant bit (MSB) as the sign indicator (0 for positive, 1 for negative) and the remaining seven bits to represent the absolute numerical magnitude. This architecture allows for a representable range from -127 to +127.The system architecture is designed using standard logic gates and digital subsystems, then modeled and validated using digital simulation software. The primary focus of the simulation is to verify correct data encoding, decoding, and the visual representation of signed integers. Additionally, the paper analyzes the inherent limitations of the sign-magnitude approach, specifically the redundant dual representation of zero (+0 and -0) and the complex hardware overhead required for arithmetic operations compared to complement-based systems. The simulation results confirm that the design accurately processes and displays 8-bit signed-magnitude values, proving its educational and functional viability for basic digital systems.